

Macro AI Hiring Report - Freudenberg Group

AI capabilities identified from investment signals

Dataset facts: 3,617 observed postings, 79 AI-core postings, and 86 AI-adjacent postings. The sections below summarize the investment_signal field for AI-core postings only. Counts are observed postings in the dataset, not confirmed hires, budget, capex, or R&D expense.

Materials Informatics and Molecular Discovery. The company is investing in machine learning, physics-based modeling, and molecular dynamics simulations to predict material behavior and molecular properties. These capabilities are applied to technical textiles, specialty lubricants, and functional molecules to accelerate the discovery of sustainable materials and mobility solutions. The objective is to enhance R&D innovation through virtual screening, chemical analysis, and materials informatics.

Advanced Product Engineering and Design Optimization. Investment is directed toward AI-driven surrogate modeling, Bayesian optimization, and the integration of AI with computational tools like FEA, CFD, and CAD. These technologies are used to optimize the design of industrial sealing systems, battery modules, and electric vehicle components. The goal is to automate engineering workflows, accelerate product development cycles, and improve the performance of high-performance material systems.

Manufacturing Process Control and Quality Inspection. The company is building capabilities in neural network-based model predictive control (NARX, QARX), real-time sensor data evaluation, and computer vision. These applications are used to regulate heavy-duty fuel cell systems, optimize injection molding cycle times, and automate the quality inspection of elastomer samples. The objective is to enhance manufacturing precision, predict product quality, and optimize chemical surface treatment processes.

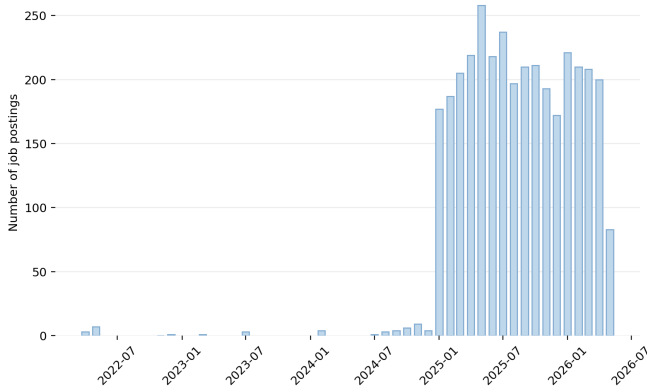
Industrial Robotics and Autonomous Systems. Investment is focused on reinforcement learning, imitation learning, and foundation models (LLM, VLM, VLA) for robotic perception and motion control. These capabilities are applied to humanoid robots to automate industrial handling and assembly tasks. The objective is to increase robotic autonomy and improve human-robot collaboration in production environments.

Enterprise Process Automation and Decision Support. The company is investing in Large Language Models (LLMs), agentic AI workflows, and Microsoft Copilot integration to automate business processes. These tools are applied to investment procurement, finance reporting, and international business collaboration. The objective is to increase operational efficiency through automated decision support, bot-driven procurement, and optimized reporting insights.

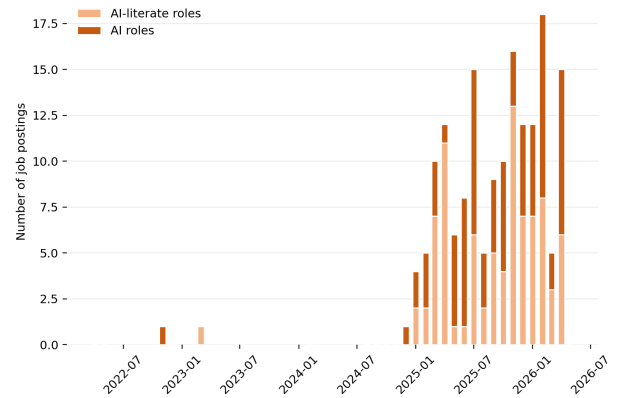
AI Infrastructure, Data Strategy, and Talent Development. Investment is being made in cloud-based data hubs, GenAIOps, and scalable AI architectures to support the deployment of machine learning models across the enterprise. This infrastructure is paired with a data management strategy specifically for R&D and laboratory automation (Lab 4.0). Additionally, the company is investing in dual bachelor's programs and internal training in data science to build a technical talent pipeline for modeling real-world hardware and software processes.

AI hiring trend

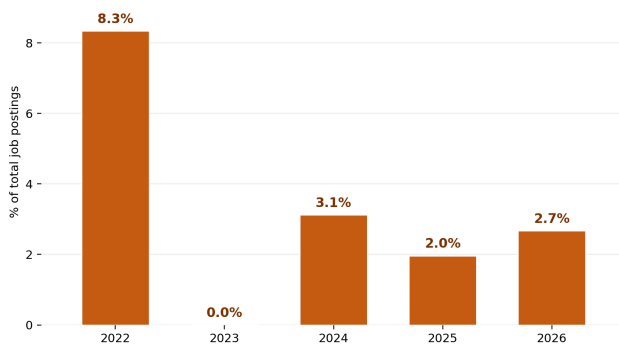
No AI roles by month



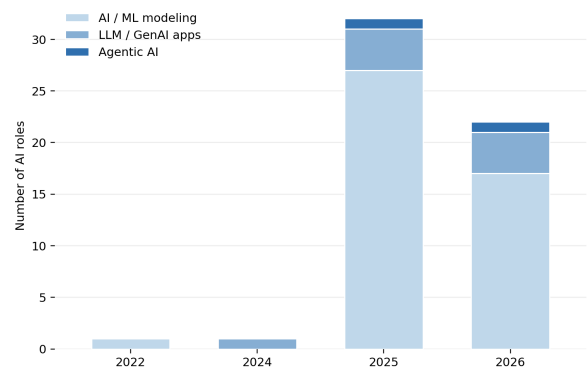
AI-related roles by month



AI roles share by year



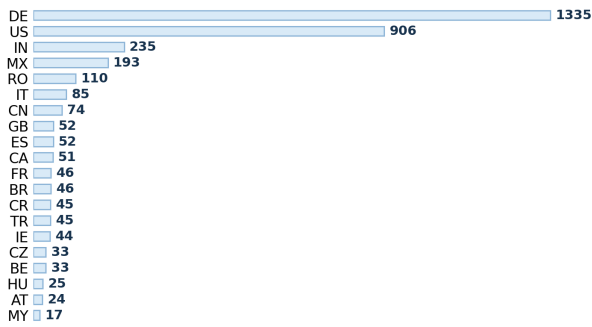
AI roles mix by year



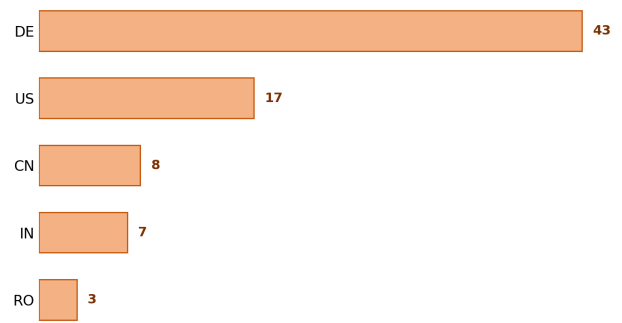
The dataset contains 3617 observed postings. AI-core postings are 79; AI-adjacent postings are 86. Years with AI-related evidence: 2022: AI-core 1/12 (8.3%), AI-adjacent 0/12; 2023: AI-core 0/5 (0.0%), AI-adjacent 1/5; 2024: AI-core 1/32 (3.1%), AI-adjacent 0/32; 2025: AI-core 51/2596 (2.0%), AI-adjacent 61/2596; 2026: AI-core 26/972 (2.7%), AI-adjacent 24/972. AI-core capability mix by year: 2022: AI / ML modeling 1; 2024: LLM / GenAI apps 1; 2025: AI / ML modeling 27, LLM / GenAI apps 4, Agentic AI 1; 2026: AI / ML modeling 17, LLM / GenAI apps 4, Agentic AI 1. Read this page as posting activity over time, not as budget, confirmed hires, or employee share.

AI hiring geography

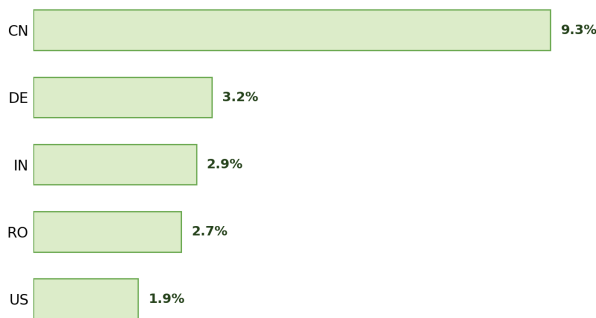
Top hiring countries



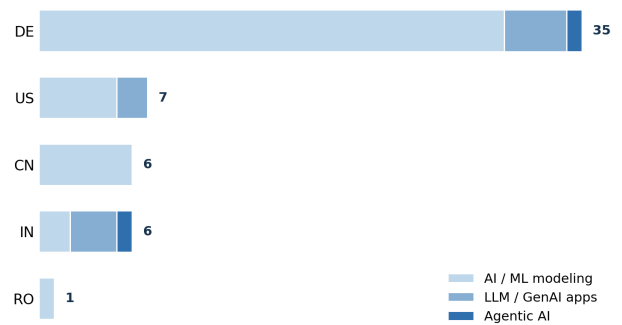
Top countries for AI roles



AI intensity by country



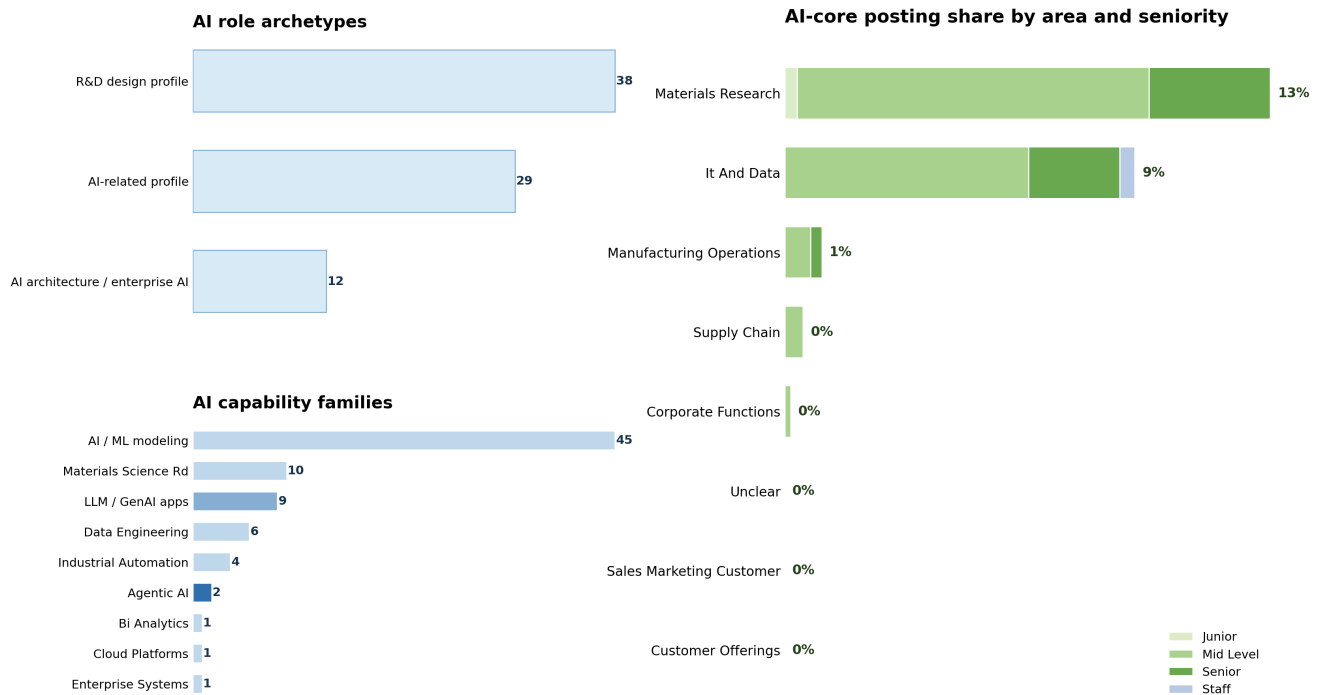
AI role mix in top AI countries



AI / ML modeling
LLM / GenAI apps
Agentic AI

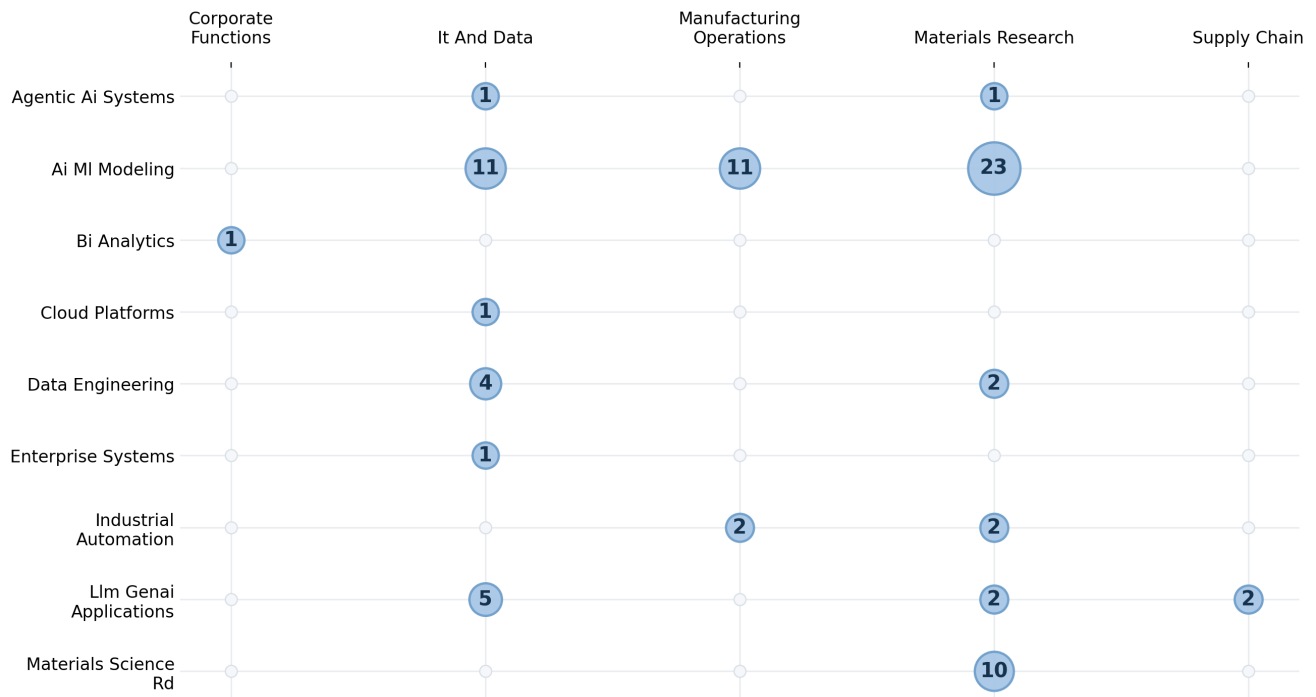
Largest overall posting countries: DE 1335; US 906; IN 235; MX 193; RO 110; IT 85. Countries with AI-core postings and their denominators: DE: 43/1335 (3.22%); US: 17/906 (1.88%); CN: 8/74 (10.81%); IN: 7/235 (2.98%); RO: 3/110 (2.73%); UNKNOWN: 1/11 (9.09%). Use the denominators when reading country percentages. A smaller country posting base can produce a higher percentage from a small number of AI-core postings.

AI staffing and seniority



This page replaces skill lists with higher-level staffing information. Role archetypes: R&D design profile: 38; AI-related profile: 29; AI architecture / enterprise AI: 12. AI capability families: AI / ML modeling: 45; Materials Science Rd: 10; LLM / GenAI apps: 9; Data Engineering: 6; Industrial Automation: 4; Agentic AI: 2; Enterprise Systems: 1; Cloud Platforms: 1; Bi Analytics: 1. Business-area denominators: Materials Research 40/306 (13.07%); It And Data 23/244 (9.43%); Manufacturing Operations 13/1302 (1.00%); Supply Chain 2/405 (0.49%); Corporate Functions 1/643 (0.16%). Seniority counts inside AI-core postings: Mid Level 57, Senior 20, Junior 1, Staff 1. Use this page to understand the type of profiles being staffed and where they sit in the organization, not to infer budget or employee share.

AI breadth across capability pockets



The matrix locates AI-core postings by capability family and business area. Capability-family counts: Ai ML Modeling 45; Materials Science Rd 10; Llm Genai Applications 9; Data Engineering 6; Industrial Automation 4; Agentic Ai Systems 2; Enterprise Systems 1; Cloud Platforms 1; Bi Analytics 1. Business-area counts: Materials Research 40; It And Data 23; Manufacturing Operations 13; Supply Chain 2; Corporate Functions 1. Nonzero cells: Ai ML Modeling in Materials Research: 23; Ai ML Modeling in It And Data: 11; Ai ML Modeling in Manufacturing Operations: 11; Materials Science Rd in Materials Research: 10; Llm Genai Applications in It And Data: 5; Data Engineering in It And Data: 4; Data Engineering in Materials Research: 2; Industrial Automation in Manufacturing Operations: 2; Industrial Automation in Materials Research: 2; Llm Genai Applications in Materials Research: 2; Llm Genai Applications in Supply Chain: 2; Agentic Ai Systems in It And Data: 1; Agentic Ai Systems in Materials Research: 1; Bi Analytics in Corporate Functions: 1; Cloud Platforms in It And Data: 1; Enterprise Systems in It And Data: 1. Read the bubbles as a map of where the observed AI-core postings sit, not as project size, budget, or strategic priority.

Appendix: brief label guide

Score 0 means no explicit AI-role signal. A posting can still be technical and remain score 0. Score 1 means AI-adjacent: AI is mentioned, but it is not the core responsibility of the role. Score 2, called AI-core in this report, means AI, ML, LLMs, GenAI, MLOps, or related AI capability is explicit and central to the role. Primary category describes the capability family, such as AI / ML modeling, LLM / GenAI applications, agentic AI systems, industrial automation, enterprise systems, or materials-science R&D. Business area describes the organizational context, such as IT and data, manufacturing operations, materials research, supply chain, customer offerings, sales and marketing, or corporate functions. Counts are observed postings in the enriched dataset. They are not confirmed hires, employee counts, budgets, capex, or R&D expense.